

SIMATSENGINEERING

ASSESSMENTTOOL2

COURSENAME:COMPUTERNETWORKS COURSE
CODE: CSA07

COURSEOUTCOMECOVERED

CO2: Configure IP addressing schemes,subnetting,and basic routing techniques using simulation tool, for efficient network design and topology.(BL3)

AssessmentToolWeightage: 30%

QUESTIONS: 5

Total Marks:50

AnnexureA

Scenario-Based

Codeof Conduct:

*I _____(Name/RegNo)certifythatthissubmissionismyoriginalworkand that I have adhered to the guidelines specified for this assessment.
Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinary action.*

Signatureofthe student:

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the **total number of possible host addresses** available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Question 1

Scenario Understanding

The Smart Campus has been assigned the IPv4 network **192.168.100.0/24**. Different departments require different subnet sizes based on the expected number of devices. Variable Length Subnet Masking (VLSM) is used to efficiently divide the address space while minimizing wastage.

Step 1: Subnet Allocation

Department	Subnet Mask	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.1	192.168.100.126	192.168.100.127	126
School of Electronics	/26	192.168.100.128	192.168.100.129	192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.193	192.168.100.222	192.168.100.223	30

Step 2: Remaining Address Space

Unused subnet:

- Network Address: **192.168.100.224/27**
- Host Range: **192.168.100.225 – 192.168.100.254**
- Broadcast Address: **192.168.100.255**

Decision and Justification

The largest subnet (/25) is assigned first, followed by /26 and /27, ensuring efficient utilization of the available IP address space. The remaining /27 block can be reserved for future expansion, guest Wi-Fi, or network management devices. This design minimizes address wastage while allowing scalability.

Question 2

Scenario Understanding

The multinational company has been allocated the private network **10.10.0.0/16**. Each department has different host requirements, so VLSM is used to allocate subnet sizes according to departmental needs.

Step 1: Subnet Details

Department	Prefix	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1	10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1	10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129	10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193	10.10.1.222	10.10.1.223	30

Remaining IP Address Space

Remaining range:

10.10.1.224 – 10.10.255.255

Decision and Justification

The address allocation satisfies current departmental requirements and leaves a significant portion of the /16 network unused. This remaining address space can support future departments, branch offices, servers, or expansion without requiring a redesign of the network.

Question 3

Scenario Understanding

The hospital is migrating from IPv4 to IPv6 to support thousands of medical devices and IoT sensors. IPv6 provides a much larger address space, simplifying future expansion.

Solution

Given Address

2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Address

2001:db8:abcd::/64

2. Expanded Address

2001:0db8:abcd:0000:0000:0000:0000/64

3. Network Prefix

2001:db8:abcd:0000::/64

4. Total Host Addresses

A /64 IPv6 subnet provides

$2^{64} = 18,446,744,073,709,551,616$ addresses

Decision and Justification

A /64 prefix is the standard IPv6 subnet size. It supports automatic address configuration (SLAAC), ensures compatibility with modern networking standards, and provides sufficient addresses for all present and future hospital devices.

Question 4

Scenario Understanding

The ISP router forwards incoming packets based on the destination IP address using its routing table. Longest-prefix matching is used to identify the correct destination network.

Routing Table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP:

192.168.2.45

Solution

Destination Subnet

192.168.2.0/24

Matching Route

192.168.2.0/24

Next-Hop Interface

The interface connected to the **192.168.2.0/24** network.

Default Route

0.0.0.0/0

Decision and Justification

Since the destination address belongs to the **192.168.2.0/24** subnet, the router forwards the packet through the interface connected to that subnet. If no matching route existed, the packet would be forwarded through the default route (0.0.0.0/0).

Question 5

Scenario Understanding

A company has divided its network into two LANs connected through a Layer-3 router. Each LAN has its own subnet and gateway, allowing efficient communication while reducing broadcast traffic.

Administration LAN

- Network Address: **192.168.10.0/26**
- Broadcast Address: **192.168.10.63**
- Host Range: **192.168.10.1 – 192.168.10.62**
- Default Gateway: **192.168.10.1**

Suggested IP Addresses

- PC1 → 192.168.10.2
- PC2 → 192.168.10.3
- PC3 → 192.168.10.4

Finance LAN

- Network Address: **192.168.10.64/26**
- Broadcast Address: **192.168.10.127**
- Host Range: **192.168.10.65 – 192.168.10.126**
- Default Gateway: **192.168.10.65**

Suggested IP Addresses

- PC1 → 192.168.10.66
- PC2 → 192.168.10.67
- PC3 → 192.168.10.68

Decision and Justification

The router gateways (**192.168.10.1** and **192.168.10.65**) enable communication within each subnet and between the two LANs. Assigning host IP addresses immediately after the gateway simplifies network management and follows standard IP addressing practices. The /26 subnet supports **62 usable hosts** per LAN, providing room for future expansion.

RUBRICSFORCALCULATION

ScenarioBased Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand